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B. E. (Sixth Semester) Examination, Nov.-Dec. 2015

(Old Scheme)

(Elect. Engg.)

FIBRE OPTICS

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note *: Attempt all questions. Part (a) of each question is compulsory & carries 2 marks. Solve any two parts from part (b), (c) and (d) & carries 7 marks each.*

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Unit-I

1. (a) What is normalized frequency of an optical fibre?

(b) Explain the vapour phase oxidation process of fibre fabrication along with the advantage and disadvantage of it.

(c) A graded index fibre with triangular profile supports the propogation of 500 modes. The core axis refractive index is 1.44 and core diameter is 75 μm if the wavelength of light propagation th' the fibre is 1.3 μm, calculate :

(i) Refractive index difference of the fibre.

(ii) Maximum diameter of the fibre core which would give single mode operation at the same wavelength.

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(d) Prove that for an optical fiber :

$$N_A = n_1 \sqrt{2\Delta}$$

$N_A \rightarrow$ Numerical Aperture

$n_1 \rightarrow$ Refractive index of core

$\Delta \rightarrow$ Relative refractive index difference

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Unit-II

2. (a) A photo diode is constructed of GaAs which has band gap energy of 1·43 eV at 300 K. Determine the cut-off wavelength.

(b) Two step index fibres exhibits following parameters :

(i) A multimode with core *RI* of 1·40 a relative refractive index of 5% and an operating wavelength of 0·95 μm.

(ii) A 10 μm core diameter single mode fibre with core *RI* 1·46 a relative refractive index difference 0·5% and operating wavelength 1·33 μm.

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Estimate critical radius of curvature of which large bending loss occurs in both case.

(c) Explain and compare the modulation process of LED & LASER diodes discussing with the help of circuit diagram.

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(d) Describe fusing splicing of optical fibers. Discuss the advantags and drawbacks of this joining techniques.

Unit-III

3. (a) What do you mean by ISI?

(b) Discuss the various mechanism responsible for attenuation in optical fibres and their remedies.

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(c) An optical fibre has attenuation resulting from rayleigh scattering 0·26 db/km at a wavelength of 1 μm. The glass has an effective temp. of 758 K. Isothermal compressibility is given by $8{\cdot}48 \times 10^{-11}$ m^2/V and photoelastic coefficient is 0·245. Determine the Refractive index of material.

(d) Write short notes on :

(i) Mode Coupling

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(ii) Dispersion shifted and dispersion flattened fibre

Unit-IV

4. (a) What do you mean by Coherent Optical detection?

(b) Write notes on :

(i) Line coding

(ii) Eye pattern

(c) Explain Link power budget and Rise time budget.

(d) Explain principle of coherent detection and thus prove that for homodyne detection.

$$I_P = \frac{2nq}{n_r}\sqrt{P_S P_L}\cos\phi$$

Unit-V

5. (a) What do you mean by FDDI?

(b) Write a brief note on Optical Amplifiers.

(c) Draw a block diagram of SONET architecture and describe its operation.

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(d) Explain wavelength division multiplexing.

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